

AI-Powered Predictive Maintenance for Industrial Equipment

Task Assignment & Roles

Defined Responsibilities for Team Members

Team Members

Alaa Mohamed Hamed Awad
Nada Ahmed Jalal
Taghreed Mohammed Abdelkader
Rowan Nasser Abbas

1. Team Member Profiles & Roles

1.1 Alaa Mohamed – Lead Engineer

Primary contributor across all project phases. Responsible for the complete data pipeline on the MIMII dataset, both predictive models (LSTM and Autoencoder), all model deployments, and MLflow integration.

Key Responsibilities:

- CMAPSS and MIMII data preprocessing and analysis
- LSTM-based RUL Prediction model development (co-lead)
- Autoencoder-based Acoustic Anomaly Detection model (sole lead)
- FastAPI deployment for all models
- MLflow integration and experiment tracking (co-lead with Nada)

1.2 Rowan Nasser – Data Analyst & Model Co-Developer

Focused on data exploration and co-development of the RUL prediction model.

Key Responsibilities:

- Data collection and exploration (CMAPSS dataset)
- RUL Prediction model development and optimization (co-lead with Alaa)

1.3 Nada Ahmed – Data Engineer & MLOps Specialist

Responsible for CMAPSS data preprocessing, MLflow integration, and continuous monitoring pipelines.

Key Responsibilities:

- CMAPSS data preprocessing and analysis
- MLflow integration (co-lead with Alaa)
- Continuous monitoring and automated retraining pipeline

1.4 Taghreed Mohammed – Data Analyst & Feature Engineer

Concentrated on data preprocessing, advanced analysis, and feature engineering for the CMAPSS dataset.

Key Responsibilities:

- CMAPSS data preprocessing, exploration, and analysis
- Feature engineering (rolling averages, degradation indicators, statistical features)

2. Task Assignment Matrix

Task	Phase	Assigned To
CMAPSS Data Collection & Exploration	Phase 1	Alaa, Taghreed, Nada, Rowan
MIMII Data Collection & Exploration	Phase 1	Alaa
CMAPSS Data Preprocessing	Phase 1	Alaa, Taghreed, Nada
MIMII Data Preprocessing & Analysis	Phase 1	Alaa
Advanced Analysis & Feature Engineering (CMAPSS)	Phase 1	Alaa, Nada, Taghreed
Data Visualization	Phase 1	Rowan, Taghreed, Alaa
LSTM RUL Prediction Model (Build & Train)	Phase 2	Rowan, Alaa
Autoencoder Anomaly Detection (Build & Train)	Phase 2	Alaa
Model Finetuning & Hyperparameter Optimization	Phase 2	Alaa, Rowan
Model Evaluation & Reporting	Phase 2	Alaa, Rowan, Nada, Taghreed
MLflow Experiment Tracking & Registry	Phase 3	Alaa, Nada
FastAPI Model Deployment	Phase 3	Alaa
Monitoring & Retraining Pipeline	Phase 3	Nada

3. RACI Summary

R = Responsible | A = Accountable | C = Consulted | I = Informed

Area	Alaa	Rowan	Nada	Taghreed
CMAPSS Preprocessing	R/A	C	R	R

Area	Alaa	Rowan	Nada	Taghreed
MIMII Preprocessing	R/A	I	I	I
RUL Prediction (LSTM)	R/A	R	C	C
Anomaly Detection (Autoencoder)	R/A	I	I	I
MLflow Integration	R/A	I	R	I
Model Deployment	R/A	I	I	I
Monitoring Pipeline	C	I	R/A	I